

Fault Diagnosis in Industrial Systems

Abstract:

Industrial systems consist of various interconnected components such as sensors, motors, pumps, valves, and control units. A fault in any component can affect the overall performance of the system, leading to production delays, increased maintenance costs, and safety risks. Therefore, early detection and diagnosis of faults are essential for ensuring reliable and efficient industrial operations.

This project presents an AI-based Fault Diagnosis System that utilizes the Breadth First Search (BFS) algorithm to identify and analyze faults in industrial systems. The system models industrial components and their relationships as a graph, where each node represents a component and edges represent connections between components. When a fault is reported, the BFS algorithm systematically explores all connected components level by level to locate the source of the fault and identify related issues.

The proposed system helps maintenance personnel quickly diagnose faults, understand fault propagation, and take corrective actions. By using BFS, the system ensures comprehensive exploration of all possible fault paths, improving fault detection accuracy and reducing diagnosis time. The project demonstrates the application of Artificial Intelligence search techniques in industrial fault management and decision support systems.

Keywords: Fault Diagnosis, Industrial Systems, Artificial Intelligence, BFS Algorithm, Graph Traversal, Predictive Maintenance.

Plan of Action:

Step 1: Understand the Problem

Study different types of faults that occur in industrial systems.

Identify the components involved in the system.

Step 2: Design the System

Represent the industrial system using a graph structure.

Consider each component as a node and each connection as an edge .

Step 3: Implement BFS Algorithm

Apply the Breadth First Search (BFS) algorithm to traverse the graph.
Check connected components level by level to find faults.

Step 4: Develop the Application

Create a simple menu-driven program.

Add options to:

Detect faults

View fault types

Display diagnosis results

Exit the program

Step 5: Test the System

Run the program with different fault scenarios.

Verify whether the faults are identified correctly.

Step 6: Analyze Results

Observe the diagnosis results.

Check the effectiveness of the BFS algorithm in fault detection.

Step 7: Documentation and Presentation

Prepare the project report.

Create a PowerPoint presentation with objectives, methodology, implementation, results, and conclusion.

Thank You